

A Field Review of Five Themes · Physiography to Harvest

Image: Topographic map of Pakistan showing mountains, Indus plain, and surrounding seas

Geography of *Pakistan*

Mountain ranges, seasonal rhythms, rainfall, irrigation, and the crops that grow from the Indus — a coordinate system for the fifth-largest South Asian country.

COURSE

Physical Geography of South Asia

Undergraduate · 200-level · Lecture 11 of 28

READING SCOPE

5 Themes · 1 Country

Mountains · Seasons · Rainfall · Irrigation · Crops

COMPANION READING

Khan, F.K. (1991) *A Geography of Pakistan*; updated with PMD and WAPDA data.

What this lecture covers — and what it leaves out.

This module is a *survey*, not a monograph. We trace **five interlocking themes** that together explain why Pakistan looks the way it does on a map — why its farms sit where they sit, why its rivers run where they run, and why its climate is one of the most extreme on Earth.

Image: Pakistan outline map with five thematic regions highlighted

THEME 01 · PHYSIOGRAPHY

Mountain Ranges

Himalayas, Karakoram, Hindu Kush, Hindu Raj, Sulaiman, Kirthar — the uplifts that frame the Indus plain.

THEME 02 · CLIMATE

Seasonal Patterns

A tropical continental rhythm: cold dry winter, scorching pre-monsoon, monsoon rains, retreating monsoon.

THEME 03 · HYDROLOGY

Rainfall System

Monsoon depressions, westerly disturbances, convective storms — and their stark spatial gradient.

THEME 04 · ENGINEERING

Irrigation System

The Indus Basin Irrigation System — the largest contiguous irrigation network on Earth.

THEME 05 · AGRICULTURE

Major Crops

Cotton, wheat, rice, sugarcane, maize — and the agro-ecological zones that determine which grows where.

DELIBERATELY OUT OF SCOPE

Demography · Politics · Geology

Population and economic geography appear only where they shape the five themes above.

Reading the deck

Slides 05–07 explore mountains; slides 08–09, seasons; 10–12, rainfall; 13–16, irrigation; 17–19, crops. Slide 20 returns to the Indus basin as a single integrated case study. Slide 22 lists the references.

A 3 × 5 map of Pakistan’s five physical themes.

Rows classify the **driving system**: the lithosphere that built the mountains; the atmosphere that delivers the rains; the engineered hydrology that carries them to fields.

Columns classify the **surface expression**: where the system shows up on a soil map, a rainfall isohyet, or a crop calendar.

The lecture visits the cells in clockwise order from R1·C1 on slide 05.

LEGEND — CELL DENSITY

- densely studied
- moderately studied
- emerging / sparse
- research gap / open*

	C1 Mountain Belt	C2 Riverine Plain	C3 Delta / Coast	C4 Arid Balochistan	C5 Northern Highlands
R 1 Lithosphere & Uplift	R1·C1 Karakoram, Himalaya 5,000–8,611 m; ongoing Indo–Eurasian convergence	R1·C2 Indus Suture Piedmont alluvium from Himalayan erosion fills the plain	R1·C3 Indus Delta Active subsidence; mangrove coast losing elevation	R1·C4 Chagai Arc Volcanic & folded ranges; Iranian suture influence	R1·C5 Hindu Kush Pamir-syntaxis cutting east–west across Afghanistan border
R 2 Atmosphere & Monsoon	R2·C1 Orographic rain Summer monsoon stalled against Siwaliks	R2·C2 Monsoon core Punjab & Sindh receive 300–700 mm Jul–Sep	R2·C3 Tropical cyclone Arabian Sea storms bring late-season rain	R2·C4 Hyper-arid <100 mm annual; convective dust storms	R2·C5 Westerlies Dec–Mar disturbances carry snow to the Hindu Kush
R 3 Hydrosphere & Crops	R3·C1 Terrace wheat Rain-fed rabi on mountain benches, NWFP	R3·C2 Canal colonies Cotton–wheat rotation under Indus Basin Irrigation System	R3·C3 Rice delta Sindh’s rice belt & saline-tolerant varieties	R3·C4 Sparse cropping <i>Open gap: drought-tolerant crops under climate stress</i>	R3·C5 Stone fruit Apricots, apples, walnuts; valley agriculture in Chitral & Hunza

Background

Pakistan at a glance: 881,913 km² of *extreme* physiography.

From the second-highest peak on Earth to a sea-level mangrove delta, almost every climate and landform class is present within a single political boundary.

Area 881,913 km² 33rd largest globally	Population ~241 million 5th largest globally	Coastline 1,046 km Arabian Sea, south	Elevation span 0 – 8,611 m K2 — Godwin-Austen
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“Pakistan presents, within a single state, a remarkable range of physical conditions — from the perpetual snow of the Karakoram to the tropical mangrove swamps of the Indus delta.”

— Fazle Karim Khan, *A Geography of Pakistan* (1991), p.1

FIVE PHYSIOGRAPHIC REGIONS

- **The Northern and Western Mountain Belt** — Himalayas, Karakoram, Hindu Kush, Pamir
- **The Balochistan Plateau** — arid folded ranges oriented east–west
- **The Potwar Plateau and Salt Range** — intermontane basins of northern Punjab
- **The Indus River Plain** — one of the world’s largest alluvial plains
- **The Indus Delta and Arabian Sea Coast** — sub-tidal mangroves and tidal creeks

Image: Cross-section diagram of Pakistan’s physiographic provinces — from the Arabian Sea coast, across the Indus plain, the Salt Range, Potwar Plateau, to the Karakoram/Hindu Kush.

Cluster 1 · Northern highlands

Five mountain systems brought *together* by plate convergence.

Image: Map showing Pakistan’s five mountain systems — Himalayas in NE, Karakoram centre–north, Hindu Kush NW, Sulaiman–Kirthar west, Salt Range in Potwar Plateau.

The mountains of Pakistan are not a single range. They are a **convergent tectonic knot** where the Indian, Eurasian, and Arabian plates meet, producing the highest peaks outside the Himalaya proper, the deepest gorges in Asia, and the most seismically active fault network in the country.

- 1

HIMALAYAS

Eastern anchor of Pakistan’s mountain belt

Western syntaxis in Azad Kashmir; Nanga Parbat (8,126 m) marks the western Himalayan terminus.
- 2

KARAKORAM

The “roof of the world” west of the Himalaya

K2 (8,611 m), Gasherbrum I (8,080 m), Broad Peak; home to the Baltoro, Biafo, and Hispar glacier systems.
- 3

HINDU KUSH

The western wall of the Karakoram corridor

Tirich Mir (7,708 m) anchors the Chitral district; separates the Kabul River basin from the Indus.
- 4

SULAIMAN & KIRTHAR

The west-Pakistan ranges guarding the Indus plain

Sulaiman (Takht-e-Sulaiman, 3,487 m); Kirthar in Sindh/Balochistan borderlands.
- 5

SALT RANGE

Pakistan’s geological showcase

Edwards-era evaporite outcrop, Potwar Plateau; world’s second-largest rock-salt deposit.

Detail A

Image: Annotated cross-section diagram — Karakoram thrust / Main Karakoram Thrust / Indus Suture.

The northern highlands.

Karakoram, Himalaya, and Hindu Kush — a triangular syntaxis.

KARAKORAM Ceiling: 8,611 m K2, the world’s second-highest peak, anchors a 600-km range shared between Gilgit-Baltistan and Xinjiang. Five of the world’s fourteen 8,000-m peaks lie entirely within Pakistan.	HIMALAYAS Ceiling: 8,126 m Nanga Parbat, the “Killer Mountain,” stands alone in the western syntaxis of Azad Kashmir. Produces the Indus’ upper gorge — one of the deepest fluvial cuts on Earth.	HINDU KUSH Ceiling: 7,708 m Tirich Mir in Chitral marks Pakistan’s NW corner. The Hindu Kush cuts east–west across the Afghan frontier and channels the westerly disturbances that snow the Karakoram in winter.
Glaciers ~7,200 incl. 60+ km Siachen	Ice reserves ~3,700 km³ Outside polar regions, near-largest	Convergence rate ~40 mm/yr Indo–Eurasian, ongoing uplift

The northern highlands are not scenic — they are **hydrologically decisive**. Every drop of meltwater the Indus carries past Tarbela Dam originates here.

Detail B

Image: Annotated map of western Pakistan mountain ranges — Salt Range in Potwar (top), Sulaiman running N–S along Punjab/ Balochistan border, Kirthar running N–S along Sindh/ Balochistan border, Chagai Hills at the SW corner.

The western ranges guard the Indus plain like a fortress wall.

South and west of the Indus plain, three mountain chains shape rainfall, river course, and provincial boundary — the Sulaiman, the Kirthar, and the Balochistan ranges proper.

SULAIMAN RANGE Western batholith - Takht-e-Sulaiman peak (3,487 m) in Balochistan - Marks the eastern edge of the Afghan–Baloch basin - Forms the western wall of the Indus plain	KIRTHAR RANGE Sindh–Baloch divide - Zardak peak (2,201 m) and Gorakh Hill - Lies along the Kirthar Fold Belt - Separates upper Sindh from Balochistan	CHAGAI & CENTRAL MAKRAN Volcanic highland - Koh-i-Sultan volcano (2,334 m) - Chagai Hills mark Balochistan’s west - Active Makran subduction zone offshore
WHY IT MATTERS These ranges determine where the monsoon rains stop (e.g. Quetta at ~100 mm vs. Jacobabad at ~110 mm above and below the Sulaiman crest) and where the Indus plain begins. They are also the principal zone of seismic hazard outside the Karakoram.		

Image: Climate classification map of Pakistan — BWh hot desert, BSh hot semi-arid, Cwa humid subtropical, Csa subtropical highland, Dsa alpine (Karakoram only).

A four-season country. Pakistani climate varies sharply by month and latitude.

Pakistan’s climate is classified as **tropical continental**, but the dominant rhythm is governed by the South Asian monsoon, not by latitude. The same year produces dense snow at Skardu in January and a 50°C heat low at Sibi in June.

WINTER · DEC – FEB

Cold & dry across the plains

- Punjab & Sindh plains: 5–15°C daytime
- Northern highlands: snow accumulation, glacier recharge
- Westerly disturbances bring 30–40% NWFP annual rainfall
- Fog & frost common in central Punjab from Dec to mid-Jan

MONSOON · JUL – SEP

Rain-bearing southwesterly flow

- Punjab, Sindh & KPK plains receive 60–80% of annual rain
- Heavy orographic rain against Siwaliks and Pir Panjal
- Indus discharge peaks August–early September

PRE-MONSOON · MAR – MAY

Hot, dry — the heat-low season

- Sibi, Jacobabad, Larkana exceed 45°C in May
- Northwest dust storms (“loo”) dominate NW Balochistan
- Cotton sowing begins in lower Sindh with canal water

POST-MONSOON / RETREATING · OCT – NOV

Withdraws & re-balances

- Monsoon retreats south east; clear skies prevail
- Rabi sowing window (wheat, lentils) opens Oct–Nov
- Brief Arabian Sea cyclone season affects Sindh coast

The PMD’s four conventional seasons

1. **Winter** (Dec–Feb)
2. **Pre-monsoon / Summer** (Mar–May)
3. **Monsoon** (Jun–Sep)
4. **Post-monsoon** (Oct–Nov)

Pakistan Meteorological Department uses these four categories rather than the conventional six- or twelve-month models.

Detail

Image: Crop calendar chart — shows the kharif (Apr–Oct) and rabi (Oct–Apr) cycles overlaid on monthly rainfall and temperature curves at Lahore.

Why **the monsoon pulse** is the single most important variable in Pakistani geography.

The seasonal rhythm is the **schedule** on which every rain-fed and many irrigated crops operate:

KHARIF SEASON · APR–OCT Monsoon-fed crops: cotton, rice, sugarcane, maize Sowing at pre-monsoon heat; growth through SW monsoon; harvest at retreating monsoon.
RABI SEASON · OCT–APR Winter crops: wheat, barley, mustard, gram, lentils Sowing on residual moisture of post-monsoon; growth on winter westerlies.
WHY IT MATTERS FOR INDUSTRY Cotton-textile & sugar industries are weather-cycle-locked: a single weak monsoon diverts billions of rupees of output.
CLIMATE TREND (CRUTEM / PMD NORMALS) Monsoon onset has shifted later by 7–10 days since 1980; summer heat extremes have increased in frequency.

Image: Mean annual rainfall isohyets — 100, 250, 500, 1,000 mm contours overlaid on a Pakistan map; note the steep east-to-west decline over Balochistan.

Pakistan’s rainfall ranges from <100 mm to >1,500 mm — within a single valley.

The country’s rainfall regime is the **most spatially skewed** in South Asia. The monsoon delivers 80% of national rain to a 30% strip; the western ranges block rain from reaching Balochistan; the North receives both monsoon moisture and winter westerlies.

1	MONSOON Southwest monsoon delivers 60–80% of rain to plains Jul–Sep; Punjab plain, Indus right-bank, KPK.
2	WESTERLY WINTER DISTURBANCES Mediterranean cyclonic storms feed the NW highlands Dec–Mar; Chitral, Dir, Swat, Murree hills.
3	CONVECTIVE THUNDERSTORMS Pre-monsoon cells: dust storms and the local “loo” Central Punjab, NW Balochistan, May–Jun.
4	TROPICAL CYCLONES Arabian Sea storms bring late-season rain Oct–Nov, May–Jun. Coastal Sindh and Makran.
5	OROGRAPHIC ENHANCEMENT Mountain-forced uplift explains local maxima Murree (~1,800 mm) vs. Sibi (~110 mm).

Detail A

Image: Mean annual rainfall isohyet map of Pakistan with city rainfalls labeled — Murree 1,800 mm, Lahore 628 mm, Quetta 100 mm, Nokkundi 40 mm.

Pakistan’s rainfall distribution, from 1,800 mm Murree to 100 mm Nokkundi.

HIGH-RAINFALL NORTH · >1,000 MM Murree, Muzaffarabad, Abbottabad, parts of Swat and Mansehra Driver: orographic uplift against Pir Panjal and Siwalik ranges during SW monsoon.
MODERATE-RAINFALL PLAINS · 300–700 MM Lahore (~628 mm), Faisalabad (~430 mm), Multan (~187 mm), Sukkur & Larkana Driver: monsoon low-pressure cells tracking NW from the Bay of Bengal.
SEMI-ARID CENTRAL · 100–300 MM Sibi (~110 mm), D.I. Khan, Sargodha, parts of southern Punjab Driver: rainshadow effect of the Sulaiman–Kirthar wall on the monsoon inflow.
HYPER-ARID WEST · <100 MM Quetta (~100 mm), Nokkundi, Dalbandin, Kharan Driver: position on the leeward side of both monsoon and most westerly disturbances.
NORTHERN HIGHLAND FRINGE · 400–1,500 MM Gilgit, Skardu, Chitral, Astor Driver: winter westerlies + summer monsoon fringe + snowmelt recharge.

Detail B

Three synoptic-scale systems, working in different seasons.

SYSTEM 1 SW Monsoon Jul–Sep · heaviest single contributor • Cross-equatorial flow from the southern Indian Ocean • Strengthened by the South Asian heat low over Sindh/Rajasthan • Bay-of-Bengal depressions track NW into Punjab & KPK • Forced ascent on the Siwaliks, Pir Panjal & Hindu Kush produces extreme localised rainfall National share: ~60–75% of annual total.	SYSTEM 2 Westerly disturbances Dec–Mar · cryospheric feed • Upper-level troughs in the subtropical westerly jet • Originate over the Mediterranean / Caspian basin • Cross Iran, Afghanistan, and into NW Pakistan • Critical for snow accumulation on the Karakoram and Hindu Kush NWFP / KPK winter-spring share: ~30–40%.	SYSTEM 3 Tropical cyclones May–Jun & Oct–Nov • Develop over the warm Arabian Sea • Mostly track into Gujarat, but coastal Sindh & Makran face direct exposure • Storm surge & heavy rain a major hazard in Thatta, Badin, Lasbela • Average 1–2 cyclones per year, with active 2019–2023 season National share: small, but locally catastrophic.
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Source: PMD Pakistan Climate Report, PMD national rainfall normals 1991–2020, AR5 / AR6 regional projections.

Where rainfall is unreliable, **engineered water** re-creates an agricultural climate.

Pakistan’s irrigation system is the largest contiguous network on Earth: roughly **83,000 km of canals**, more than 1.6 million tubewells, and two of the world’s largest earth-fill dams (Tarbela, Mangla). It is the *engineered replacement* for the rain that geography denies.

SCALE ~14.5 million hectares irrigated More than 80% of Pakistan’s agricultural output depends on irrigation. Rain-fed farming covers only ~18% of cultivated area, mostly in NWFP and northern Punjab.	CANAL LENGTH ~83,000 km of main canals British-built 19th-century system extended under the Indus Waters Treaty (1960): Lower Bari Doab, Lower Chenab, Lower Jhelum are flagship canals.
TUBEWELLS ~1.6 million operational Private tubewells dominate; many run on subsidised diesel or grid electricity. Groundwater now accounts for nearly half of all irrigation withdrawals.	DAMS Tarbela, Mangla — large storage Tarbela (14.3 km³), Mangla (9.3 km³). Diamer-Bhasha under construction; Kalabagh is the proposed but contested third storage project.

Detail A

Built under colonial gravity, transformed by the Indus Waters Treaty.

The IBIS dates to the 1859 Lower Bari Doab Canal. After partition in 1947, the headworks of the upper Indus were lost to India; the 1960 Indus Waters Treaty — brokered by the World Bank — re-engineered the system over 20 years to redirect western-river flows (Jhelum, Chenab, Indus) to former eastern-river canals.

CANAL COLONIES (BRITISH ERA)

Reclaimed the doabs

- Lower Bari Doab (Khanki) — 1859
- Lower Chenab (Khanki) — 1892
- Lower Jhelum — 1901
- Upper Sirhindia & Sidhnai — early 20th c.
- These turned the doabs into Pakistan’s cotton–wheat belt

INDUS WATERS TREATY (1960)

Funded by the World Bank

- India: Ravi, Beas, Sutlej (eastern rivers)
- Pakistan: Indus, Jhelum, Chenab (western rivers)
- \$1.7 billion replacement works: Mangla Dam, Tarbela Dam, eight inter-river link canals
- Completed in 1975 — the largest hydraulic programme in the 20th century

MODERN EXTENSIONS (1975–2026)

Greater Thal, Indus Right Bank

- Greater Thal Canal (1981) opened the Dera Ghazi Khan doab
- Kachhi Canal extension (2017) in Balochistan
- Diamer-Bhasha Dam under construction
- Moho-Jalwana, Raineer canals extend Sindh coverage

Detail B

Image: Schematic of the Indus Basin Irrigation System — Tarbela & Mangla reservoirs, link canals (Jhelum–Chenab, Chenab–Ravi), trimmu/sidhnai barrages, into Nara–Panjnad tail.

Three-tier hierarchy: **main canal → distributary → watercourse.**

Water flows by gravity (former colonial canals designed around slope) and pump (modern tubewells) through a pyramid of channels. At the base of the pyramid sits the farmer’s *watercourse* — the small earthen ditch that connects the system to the field.

MAIN CANALS Trunk lines: Upper Bari Doab, Lower Chenab, Lower Jhelum, Sidhnai, Nara, Panjnad Capacity from 200 to >500 m³/sec; rim-stations regulated by WAPDA.
DISTRIBUTARIES & MINORS ~6,000 distributaries branching to ~135,000 watercourses Maintained by provincial Irrigation Departments (Punjab, Sindh, KPK, Balochistan).
TUBEWELLS Private submersible pumps fill the groundwater gap By 2020, groundwater withdrawal ~50% of total irrigation supply; much is saline.
REFORM AND MANAGEMENT TRANSFER Area Water Boards (Sindh), Farmer Organisations (Punjab) 1994–2000 reforms devolved irrigation management to farmer organisations — an on-going institutional shift.

Headline statistic

3,000 mm

Highest annual irrigation-depth (water applied per hectare) in the lower Indus delta — a figure that signals both water abundance *and* salinity risk (FAO/WAPDA).

Controversy · diverging evidence

Image: Side-by-side maps — (left) water-table depth 2020, (right) root-zone salinity map 2020 — showing the puzzling divergence.

The same canals that created the green revolution created a slow-onset land crisis.

A classic divergent-evidence slide: official WAPDA figures say waterlogging is on the decline; provincial Department-of-Agriculture figures say salinity is on the increase. Both are correct — they describe **different processes**.

GROUP 01 · INDEPENDENT WAPDA / SCARP Programme Salinity Control & Reclamation Projects • Claim. Waterlogged area fell from 2.1 Mha (1978) to ~0.9 Mha (2020) • Tile drains installed under SCARP I/II; tubewells lowered shallow water tables • Ondo, LBOD drainage projects in Sindh • Source: WAPDA National Water Management Strategy	GROUP 02 · INDEPENDENT Pakistan Council of Research in Water Resources PCRWR provincial surveys • Claim. ~4.5 Mha (of 14.5 Mha irrigated) affected by salinity, rising • Most pronounced in southern Sindh and southern Punjab doabs • Yields collapse on saline patches by 20–60% • Source: PCRWR National Water Quality Monitoring Reports
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Reconciliation — why both are right

Waterlogging (lateral rise of the water table) has declined since the 1970s; **root-zone salinity** (vertical salt accumulation) has accumulated over a century of irrigation. Pakistan’s surface has dried, but the *soil* has become saltier — a divergence driven by irrigation history, not conflicting measurements.

Five crops carry Pakistan’s agriculture — and one rotation pattern carries them all.

Cotton and **wheat** dominate sown area and export earnings. **Rice** and **sugarcane** are the second tier — both water-intensive. **Maize** is a high-yielding substitute for fodder and starch. Together, the five crops drive ~80% of agricultural output.

The cotton–wheat rotation

Sown in sequence on the same fields of southern Punjab and lower Sindh: cotton (Apr–Nov) followed by wheat (Nov–Apr). The country’s signature cropping pattern.

CROP 01 Cotton <i>Kharif · Gossypium</i> ~2.0 Mha sown - Punjab & Sindh irrigated belt ~9.0 Mt bales (2023–24) 5th-largest producer globally - Backbone of textile exports	CROP 02 Wheat <i>Rabi · Triticum</i> ~9.0 Mha sown ~28 Mt grain (2023–24) - Punjab plain dominant - Staple food security crop ~74% of rabi sowing	CROP 03 Rice <i>Kharif · Oryza</i> ~3.0 Mha sown - Sindh & Punjab belt ~9.0 Mt milled - Basmati export — premium - Largest water user per hectare	CROP 04 Sugarcane <i>Kharif · Saccharum</i> ~1.2 Mha sown - Punjab lower belts ~80 Mt cane (2023–24) ~75 Mt sugar refining 5th-largest producer	CROP 05 Maize <i>Kharif · Zea mays</i> ~1.6 Mha sown - KP and Punjab hybrid ~11 Mt grain - Fodder + starch - Rising rapidly in KPK
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Image: Photographic collage of Pakistan’s five major crops — cotton bolls, wheat harvest, rice paddy, sugarcane, maize.

Detail A

Image: Crop calendar table — cotton/wheat/rice/sugarcane/maize with sowing, growing, harvest windows across 12 months.

Water demand differentiates the five crops — and **constrains geography.**

Rice requires 1,500–2,000 mm of water per season; sugarcane 1,500–2,500 mm; cotton and maize 600–900 mm; wheat 400–500 mm. This gradient explains why **irrigation maps and crop maps are nearly identical** across the Indus plain.

COTTON — KHARIF

Apr sow · Oct–Nov pick

- Sown April; first pick Aug–Sep (early cotton)
- Heat: G. hirsutum needs day temperatures >30°C
- Water: 700–900 mm; most is applied by furrow irrigation
- Best in central Punjab, southern Sindh

WHEAT — RABI

Nov sow · Apr harvest

- Sown on canal residual moisture, irrigated 3–5 times
- Cool-season (10–15°C) vegetative phase
- Water: 400–500 mm; relatively drought-tolerant
- Northern Punjab rain-fed stretches add ~1 Mha

RICE — KHARIF

May sow · Oct harvest

- Transplanted June; standing water 50–100 mm in fields
- Water: 1,500–2,000 mm — the largest consumer
- Restricted to Sindh (Larkana, Jacobabad) and Punjab (Gujranwala, Hafizabad)
- Climate change is squeezing rice eligibility zone

SUGARCANE & MAIZE

Kharif perennial and cereal

- Sugarcane: 14–16 month crop — sown Feb–Mar, harvested Oct–Apr
- Water: 1,500–2,500 mm — second-largest consumer
- Maize: kharif (Punjab) and spring (KPK, harvest Jul)
- Maize is the fastest-expanding acreage crop since 2010

Detail B

Pakistan’s **crop geography** maps onto its climate geography.

Zone	Climate	Dominant crop	Secondary
Indus Delta (Sindh south)	Tropical arid, 100–200 mm rain	Rice (IRRI varieties)	Sugarcane on perennial fields
Lower Indus (Sukkur, Larkana)	Subtropical arid, 100–300 mm	Cotton (G. hirsutum & BT)	Wheat on rotation
Punjab irrigated plain	Semi-arid subtropical, 300–700 mm	Wheat (rabi) · Cotton (kharif)	Sugarcane, maize, oilseeds
KPK & foothills	Subtropical highland, 400–1,200 mm	Maize (kharif) · Wheat (rabi)	Sugarcane in plains, tobacco
Northern highlands	Cold alpine / sub-alpine, 200–800 mm	Wheat, barley, potato	Apricot, apple, walnut
Balochistan (Quetta, Nasirabad)	Arid / semi-arid, 80–300 mm	Wheat (rain-fed rabi)	Cumin, onion, fodder

Source: Pakistan Bureau of Statistics, Agricultural Statistics of Pakistan 2022–23; FAO Agro-Ecological Zones for Pakistan (1993, revised 2018).

Case study · analysis

A single **basin** glues the five themes together.

The Indus basin is the country’s connective tissue. Rain falls on Karakoram glaciers; the melt becomes the Indus River; the river feeds the irrigation canals; the canals drive the cotton–wheat rotation; the rotation sustains 60% of Pakistan’s employment. The five themes are not separate chapters — they are the same chapter read at different points.



One sentence

“*The Indus is not a Pakistani river in a Pakistani country. Pakistan is, in a real sense, a country made by the Indus.*”

— paraphrased after Fazle Karim Khan, 1991

In numbers

3,180 km — Indus river length

1.1 M km² — basin area

~160 km³ — mean annual flow

90% — share of Pakistan’s irrigation supply from Indus system

Three tensions define Pakistan’s physical geography — and **climate change will reshape each.**

TENSION 01

Glacier loss vs. storage

Karakoram glaciers act as natural reservoirs — the long-run loss of ice mass threatens Indus baseflow. New storage (Diamer-Bhasha) is part of the official response.

TENSION 02

Cotton–wheat expansion vs. salinity

Cotton has spread south into marginal Sindh lands. Yields are dropping as soil salt accumulates. Saline-tolerant cultivars are the contested adaptation strategy.

TENSION 03

Urban water vs. irrigation

Karachi, Lahore, and Islamabad compete with farms for the same Indus-derived water. Urban demand, already ~10% of withdrawals, will rise to ~25% by 2050.

Falsifiable forward bets

Bet 1 · Cryosphere

If Hindu Kush winter snowfall falls below the 1991–2020 mean by >15% in two consecutive winters, Tarbela storage will not refill to design capacity in 2027/28. A testable cryospheric-trigger hypothesis.

Bet 2 · Salinity

If saline-tolerant cotton cultivars (e.g. NIAB-78) are planted on >500,000 ha in Sindh by 2030, the regional salinity trend will plateau or invert. Currently the trend is upward.

Bet 3 · Crops

If rice acreage in lower Sindh falls below 0.6 Mha by 2030 (down from ~1.4 Mha today), climate-driven water-scarcity will have replaced economic factors as the dominant driver. Currently water is not the primary driver.

Selected references · data anchored, classroom-friendly.

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